

Muhammad Usman

AI/ML Engineer | LLMs, RAG & AI Systems | Backend & Product Engineering

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🐙 github.com/nccsuzzi 🌐 [Portfolio](#) 📍 Islamabad 44000, ICT, Pakistan

Summary

AI/ML Engineer with **1.5+ years of production experience** building and shipping **LLM systems, RAG pipelines, AI agents, and computer vision applications** across defence, enterprise, SaaS, and startup environments. Experienced in taking AI products from architecture to production across backend, cloud, and application layers. Proven track record of reducing operational overhead, cutting latency, and delivering measurable business impact through intelligent automation. Published in **Springer** and author of an open-source Python package on PyPI.

Education

Bachelor's Degree in Computer Sciences

Air University, Islamabad, Pakistan

09/2021 – 07/2025

GPA: 3.10 / 4.00

Skills

AI/ML & AI Systems

NLP, PyTorch, TensorFlow, Scikit-learn, OpenCV, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Fine-Tuning, Semantic Search, AI Agents, Pinecone, FAISS, ChromaDB, OpenSearch, Vector Search, Multi-Agent Systems, Data Pipelines, LiteLLM, Langfuse, Mediapipe

Backend & MLOps/Cloud

FastAPI, Flask, Django, REST APIs, PostgreSQL, SQLAlchemy, AWS S3, AWS EC2, AWS Lambda, AWS Bedrock, GCP Cloud Run, GCP Compute Engine, Docker, Terraform, MLflow, Jenkins, Redis, Prometheus, Grafana, Sentry

Soft Skills

Technical Writing & Documentation, Stakeholder Communication, Analytical Problem-Solving, Cross-Functional Team Collaboration, Research & Experimentation, Agile Workflows

Experience

Software Engineer

8x Social

09/2026 – Present

Remote, United States

- Build and ship **AI-native products end-to-end**, spanning LLM integrations, backend services, application architecture, and cloud infrastructure in a fast-paced startup environment.
- Design and implement production **AI workflows and intelligent automation systems**, integrating LLMs, retrieval, structured generation, and external AI services into user-facing products.
- Own features from **architecture through deployment**, working across **Next.js, FastAPI, Supabase, GCP, and Python** while contributing to product decisions, security, accessibility, and developer tooling.

AI Engineer

Phebsoft

09/2025 – 08/2026

Islamabad, Pakistan

Client Partnership: Gitwork

- Architected **multi-agent systems** capable of autonomously performing tasks such as data scraping, information retrieval, and workflow automation.
- Developed **RAG pipelines with vector embeddings** to enable contextual Q&A across large product documentation datasets.
- Designed scalable AI APIs using **FastAPI**, delivering real-time inference.
- Deployed Docker-based AI microservices on **GCP Compute Engine**, streamlining integration between ML models and production backend systems.

AI Engineer

NASTP

02/2025 – 08/2025

Rawalpindi, Pakistan

- Processed and curated **100k+ digital forensic records** for AI model training, fine-tuning, and evaluation.

- Fine-tuned **transformer-based language models** for document classification and semantic retrieval using **AWS SageMaker**, improving categorisation accuracy by **18%**.
- Delivered **5+ RESTful AI APIs** covering OCR, speech, and voice processing using Python and FastAPI.
- Optimised model serving pipelines achieving **<400 ms inference latency** across production endpoints.

Projects

Afterdesk AI [\[Production Link\]](#)

Stack: *OpenCV, Python, FastAPI, PyMuPDF*

- Built an AI-powered document automation pipeline using **OpenCV** and computer vision to analyze arbitrary PDF templates, detect layout elements and form fields, and automatically map structured data to the appropriate regions.
- Developed a **FastAPI**-based processing service that combines document extraction, template mapping, and automated PDF generation to support dynamic document creation across multiple template formats.

Syscore AI [\[Internal\]](#)

Stack: *AWS Bedrock, AWS S3, FastAPI, Python, SYSPRO ERP*

- Developed an **AI-powered middleware for SYSPRO ERP** using **AWS Bedrock** for foundation model orchestration and **AWS S3** for XML record storage, enabling natural language and multimodal inputs.
- Achieved **10× faster data retrieval** and **~99% reduction** in ERP request formatting errors by streamlining input parsing and validation.

Speakify AI [\[Production Link\]](#)

Stack: *FastAPI, PostgreSQL, Cohere, Pinecone, LiteLLM, Langfuse, Stripe, DocuSign*

- Engineered an asynchronous **FastAPI + PostgreSQL** backend across **15+ API domains** for a B2B SaaS marketplace handling the end-to-end speaker booking lifecycle.
- Implemented a **RAG matching engine** using **Cohere embeddings** and **Pinecone** with Langfuse observability; integrated **Stripe** and **DocuSign** for commission splits and contracting.
- Transformed manual workflows into a zero-touch revenue pipeline managing **360+ user profiles** and live bookings.

Kernel-Global [\[Production Link\]](#)

Stack: *Python, FastAPI, Azure OpenAI, MySQL, PyMuPDF, python-docx, Docker, Pytest, Locust*

- Architected an asynchronous **GenAI microservice** using **FastAPI**, **AsyncIO**, and **Azure OpenAI** to automate candidate assessment, recruitment workflows, and candidate support.
- Built structured document-processing pipelines using **PyMuPDF** and **python-docx** and a conversational AI agent with **MySQL context retrieval**, reducing token usage and API latency by **~35%** through dynamic conversation summarization.
- Implemented **LLM security guardrails** for prompt-injection and jailbreak detection, backed by **40+ unit, integration, and security tests** and Locust-based concurrency testing.

AutoML Platform [\[Internal\]](#)

Stack: *Python, MLflow, Docker, FastAPI, Scikit-learn, Prometheus, Grafana*

- Delivered a self-service **MLOps platform** automating the model lifecycle across training, evaluation, deployment, and monitoring for multiple ML frameworks.
- Reduced average model deployment time by **~60%** and monitoring setup overhead by **~70%** through automated pipelines and Prometheus/Grafana observability.

Key Achievements

📄 Publication

View-Invariant Person Re-Identification in Camera Networks by Pose-Aware AE-GAN and Attention-Driven Feature Fusion

Published in **The Visual Computer (Springer)** DOI: doi.org/10.1007/s00371-025-04300-1

📦 Open Source

Conceived and released **ctxintel**, an open-source Python package for LLM context optimisation. Implements a 6-stage deterministic pipeline (ranking, extraction, memory, compression, optimisation) using TF-IDF, spaCy NER, and extractive summarisation with zero external API calls.

PyPI: pypi.org/project/ctxintel